

1(a)(i)	any two from: 1 (fruits are) yellow and round ; 2 fruits do not resemble either parent ; 3 fruits resemble Floridor (variety) ;	2																									
1(a)(ii)	gametes correctly entered ; genotypes correctly worked out ; phenotypes match genotypes ; <i>ratio of fruit phenotypes</i> 9 yellow round : 3 yellow long : 3 green round : 1 green long ; <table><tr><td></td><td>AB</td><td>Ab</td><td>aB</td><td>ab</td></tr><tr><td>AB</td><td>AABB yellow round</td><td>AABb yellow round</td><td>AaBB yellow round</td><td>AaBb yellow round</td></tr><tr><td>Ab</td><td>AABb yellow round</td><td>AAbb yellow long</td><td>AaBb yellow round</td><td>Aabb yellow long</td></tr><tr><td>aB</td><td>AaBB yellow round</td><td>AaBb yellow round</td><td>aaBB green round</td><td>aaBb green round</td></tr><tr><td>ab</td><td>AaBb yellow round</td><td>Aabb yellow long</td><td>aaBb green round</td><td>aabb green long</td></tr></table>		AB	Ab	aB	ab	AB	AABB yellow round	AABb yellow round	AaBB yellow round	AaBb yellow round	Ab	AABb yellow round	AAbb yellow long	AaBb yellow round	Aabb yellow long	aB	AaBB yellow round	AaBb yellow round	aaBB green round	aaBb green round	ab	AaBb yellow round	Aabb yellow long	aaBb green round	aabb green long	4
	AB	Ab	aB	ab																							
AB	AABB yellow round	AABb yellow round	AaBB yellow round	AaBb yellow round																							
Ab	AABb yellow round	AAbb yellow long	AaBb yellow round	Aabb yellow long																							
aB	AaBB yellow round	AaBb yellow round	aaBB green round	aaBb green round																							
ab	AaBb yellow round	Aabb yellow long	aaBb green round	aabb green long																							

Question	Answer	Marks
1(b)	any three from: 1 meiosis, cannot occur / is disrupted ; 2 33 / odd number of, chromosomes ; 3 three copies of each (homologous) chromosome ; 4 chromosomes cannot (all), pair up / form bivalents / undergo synapsis ; 5 in prophase 1 ; 6 gametes / ovules (pollen), not formed ;	3
1(c)(i)	different genera / chromosome numbers different (22 and 40) / offspring would be sterile ;	1
1(c)(ii)	any two from: 1 genetic engineering / genetic modification / recombinant DNA technology ; 2 use restriction, enzyme / endonuclease, to cut <i>C. ecuadorensis</i>, genome / DNA ; 3 use, vector / plasmid / <i>A. tumefaciens</i> ; 4 to insert WMV-resistance gene into watermelon (cells) ;	2